

CONCLUSION ON PESTICIDE PEER REVIEW

Conclusion on the peer review of the pesticide risk assessment of the active substances *Beauveria bassiana* strains ATCC-74040 and GHA¹

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ABSTRACT

The conclusions of the European Food Safety Authority (EFSA) following the peer review of the initial risk assessments carried out by the competent authority of the rapporteur Member State Germany, for the pesticide active substances *Beauveria bassiana* strains ATCC-74040 and GHA are reported. The context of the peer review was that required by Commission Regulation (EC) No 2229/2004, as amended by Commission Regulation (EC) No 1095/2007 and Commission Regulation (EU) No 114/2010. The conclusions were reached on the basis of the evaluation of the representative uses of *Beauveria bassiana* strains ATCC-74040 and GHA as an insecticide on tomatoes for strain ATCC-74040 and on tomatoes, cucumbers and ornamentals for strain GHA. The reliable endpoints concluded as being appropriate for use in regulatory risk assessment, derived from the available studies and literature in the dossier peer reviewed, are presented. Missing information identified as being required by the regulatory framework is listed. Concerns are identified.

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KEY WORDS

Beauveria bassiana strains ATCC-74040 and GHA, peer review, risk assessment, pesticide, insecticide

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SUMMARY

Beauveria bassiana strains ATCC-74040 and GHA is one of the 295 substances of the fourth stage of the review programme covered by Commission Regulation (EC) No 2229/2004, as amended by Commission Regulation (EC) No 1095/2007.

Beauveria bassiana strains ATCC-74040 and GHA were included in Annex I to Directive 91/414/EEC on 1 May 2009 pursuant to Article 24b of the Regulation (EC) No 2229/2004 (hereinafter referred to as ‘the Regulation’) and have subsequently been deemed to be approved under Regulation (EC) No 1107/2009, in accordance with Commission Implementing Regulation (EU) No 540/2011, as amended by Commission Implementing Regulation (EU) No 541/2011. In accordance with Article 25a of the Regulation, as amended by Commission Regulation (EU) No 114/2010, the European Food Safety Authority (EFSA) is required to deliver by 31 December 2012 its view on the draft review reports submitted by the European Commission in accordance with Article 25(1) of the Regulation. This review report was established as a result of the initial evaluation provided by the designated rapporteur Member State in the Draft Assessment Report (DAR). The EFSA therefore organised a peer review of the DAR. The conclusions of the peer review are set out in this report.

Germany being the designated rapporteur Member State submitted the DARs on *Beauveria bassiana* strains ATCC-74040 and GHA in accordance with the provisions of Article 22(1) of the Regulation, which were received by the EFSA on 3 December 2007. The peer review was initiated on 22 April 2008 by dispatching the DARs to the notifiers (Intrachem Bio Italia S.p.A. for strain ATCC-74040 and Mycotech Europe Ltd. for strain GHA) and on 11 June 2008 to the Member States for consultations and comments. Following consideration of the comments received on the DARs, it was concluded that EFSA should conduct a focused peer review in the area of ecotoxicology and deliver its conclusions on *Beauveria bassiana* strains ATCC-74040 and GHA.

The conclusions laid down in this report were reached on the basis of the evaluation of the representative uses of *Beauveria bassiana* strains ATCC-74040 and GHA as an insecticide on tomatoes for strain ATCC-74040 and on tomatoes, cucumbers and ornamentals for strain GHA, as proposed by the notifiers. Full details of the representative uses can be found in Appendix A to this report.

In the area of identity of the microorganism, biological properties, physical and technical properties and methods of analysis, data gaps were identified for both strains and both formulations.

The operator, worker and bystander risk assessment cannot be finalised due to the lack of information on toxins/secondary metabolites production.

The consumer risk assessment cannot be finalised until the toxins/secondary metabolite issue is addressed.

The information on fate and behaviour in the environment was not sufficient to characterise the competitiveness/persistence of *B. bassiana* strains ATCC-74040 and GHA in soil in the context of the representative uses assessed and therefore data gaps were identified. Satisfactory information is not available to demonstrate that, under the conditions of use, any toxins/secondary metabolites produced by *Beauveria bassiana* strains ATCC-74040 and GHA will not occur in the environmental compartments in concentrations considerably higher than under natural conditions. Further data on the persistence, transformation and mobility of these compounds might be needed in order to assess the potential level of environmental exposure including groundwater.

***Beauveria bassiana* strain ATCC-74040:** From infectivity and pathogenicity a low risk to birds, mammals, fish, algae, earthworms, soil micro-organisms and non-target plants was concluded. A data gap was identified to address the risk from infectivity and pathogenicity to aquatic invertebrates and pollinators for both the field and greenhouse uses. Data gaps were concluded to address the risk to

honey bee brood and the risk to non-target arthropods infectivity and pathogenicity for the field use. A data gap was concluded for information to address the risk to sewage treatment organisms for the greenhouse use. Pending the data gaps for identification of secondary metabolites/toxins a risk assessment for non-target organisms may be required for both the field and greenhouse uses.

***Beauveria bassiana* strain GHA:** From infectivity and pathogenicity a low risk to birds, mammals, fish, aquatic invertebrates, algae, and earthworms was concluded. Data gaps were identified to address the risk to pollinators and sewage treatment organisms. Pending the data gap to finalise the soil exposure assessment it may be necessary to address the risk to soil-dwelling non-target arthropods and soil microorganisms. Pending the data gaps for identification of secondary metabolites/toxins a risk assessment for non-target organisms may be required.

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BACKGROUND

Beauveria bassiana strains ATCC-74040 and GHA is one of the 295 substances of the fourth stage of the review programme covered by Commission Regulation (EC) No 2229/2004³, as amended by Commission Regulation (EC) No 1095/2007⁴.

Beauveria bassiana strains ATCC-74040 and GHA was included in Annex I to Directive 91/414/EEC⁵ on 1 May 2009 pursuant to Article 24b of the Regulation (EC) No 2229/2004 (hereinafter referred to as ‘the Regulation’) and has subsequently been deemed to be approved under Regulation (EC) No 1107/2009⁶, in accordance with Commission Implementing Regulation (EU) No 540/2011⁷, as amended by Commission Implementing Regulation (EU) No 541/2011⁸. In accordance with Article 25a of the Regulation, as amended by Commission Regulation (EU) No 114/2010⁹ the European Food Safety Authority (EFSA) is required to deliver by 31 December 2012 its view on the draft review reports submitted by the European Commission in accordance with Article 25(1) of the Regulation (European Commission, 2008a and European Commission, 2008b). This review report was established as a result of the initial evaluation provided by the designated rapporteur Member State in the Draft Assessment Report (DAR). The EFSA therefore organised a peer review of the DAR. The conclusions of the peer review are set out in this report.

Germany being the designated rapporteur Member State submitted the DARs on *Beauveria bassiana* strains ATCC-74040 and GHA in accordance with the provisions of Article 22(1) of the Regulation, which were received by the EFSA on 3 December 2007 (Germany, 2007a and Germany, 2007b). The peer review was initiated on 22 April 2008 by dispatching the DARs to the notifiers (Intrachem Bio Italia S.p.A. for strain ATCC-74040 and Mycotech Europe Ltd. for strain GHA) and on 11 June 2008 to the Member States for consultations and comments. In addition, the EFSA conducted a public consultation on the DAR. The comments received were collated by the EFSA and forwarded to the RMS for compilation and evaluation in the format of a Reporting Table. The notifiers were invited to respond to the comments in column 3 of the Reporting Table. The comments were evaluated by the RMS in column 3 of the Reporting Table.

The scope of the peer review was considered in a telephone conference between the EFSA, the RMS, and the European Commission on 16 December 2011. On the basis of the comments received and the RMS’ evaluation thereof it was concluded that the EFSA should organise a consultation with Member State experts in the area of ecotoxicology.

The outcome of the telephone conference, together with EFSA’s further consideration of the comments is reflected in the conclusions set out in column 4 of the Reporting Table. All points that

³ Commission Regulation (EC) No 2229/2004 of 3 December 2004 laying down further detailed rules for the implementation of the fourth stage of the programme of work referred to in Article 8(2) of Council Directive 91/414/EEC. OJ L 379, 24.12.2004, p.13-63.

⁴ Commission Regulation (EC) No 1095/2007 of 20 September 2007 amending Regulation (EC) No 1490/2002 laying down further detailed rules for the implementation of the third stage of the programme of work referred to in Article 8(2) of Council Directive 91/414/EEC and Regulation (EC) No 2229/2004 laying down further detailed rules for the implementation of the fourth stage of the programme of work referred to in Article 8(2) of Council Directive 91/414/EEC. OJ L 246, 21.9.2007, p.19-28.

⁵ Council Directive 91/414/EEC of 15 July 1991 concerning the placing of plant protection products on the market. OJ L 230, 19.8.1991, p. 1-32, as last amended.

⁶ Regulation (EC) No 1107/2009 of the European Parliament and of the Council of 21 October 2009 concerning the placing of plant protection products on the market and repealing Council Directives 79/117/EEC and 91/414/EEC. OJ L 309, 24.11.2009, p.1-50.

⁷ Commission Implementing Regulation (EU) No 540/2011 of 25 May 2011 implementing Regulation (EC) No 1107/2009 of the European Parliament and of the Council as regards the list of approved active substances. OJ L 153, 11.6.2011, p.1-186

⁸ Commission Implementing Regulation (EU) No 541/2011 of 1 June 2011 amending Implementing Regulation (EU) No 540/2011 implementing Regulation (EC) No 1107/2009 of the European Parliament and of the Council as regards the list of approved active substances. OJ L 153, 11.6.2011, p.187-188.

⁹ Commission Regulation (EU) No 114/2010 of 9 February 2010 amending Regulation (EC) No 2229/2004 as regards the time period granted to EFSA for the delivery of its view on the draft review reports concerning the active substances for which there are clear indications that they do not have any harmful effects. OJ L 37, 10.2.2010, p.12.

were identified as unresolved at the end of the comment evaluation phase and which required further consideration, including those issues to be considered in consultation with Member State experts, and additional information to be submitted by the notifiers, were compiled by the EFSA in the format of an Evaluation Table.

The conclusions arising from the consideration by the EFSA, and as appropriate by the RMS, of the points identified in the Evaluation Table, together with the outcome of the expert discussions where these took place, were reported in the final column of the Evaluation Table.

A final consultation on the conclusions arising from the peer review of the risk assessment took place with Member States via a written procedure in November 2012.

This conclusion report summarises the outcome of the peer review of the risk assessment on the active substance and the representative formulation evaluated on the basis of the representative uses as an insecticide on tomatoes for strain ATCC-74040 and on tomatoes, cucumbers and ornamentals for strain GHA, as proposed by the notifiers. A list of the relevant end points for the active substance as well as the formulation is provided in Appendix A. In addition, a key supporting document to this conclusion is the Peer Review Report, which is a compilation of the documentation developed to evaluate and address all issues raised in the peer review, from the initial commenting phase to the conclusion. The Peer Review Report (EFSA, 2012) comprises the following documents, in which all views expressed during the course of the peer review, including minority views, can be found:

- the comments received on the DAR,
- the Reporting Tables (13 December 2011),
- the Evaluation Tables (16 November 2012),
- the reports of the scientific consultation with Member State experts (where relevant),
- the comments received on the assessment of the points of clarification (where relevant),
- the comments received on the draft EFSA conclusion.

Given the importance of the DARs including their addenda (compiled versions of May 2012 containing all individually submitted addenda (Germany, 2012a and Germany, 2012b)) and the Peer Review Report, both documents are considered respectively as background documents A and B to this conclusion.

THE ACTIVE SUBSTANCE AND THE FORMULATED PRODUCT

THE IDENTITY OF THE MICROORGANISM AND THE PROPERTIES OF THE FORMULATED PRODUCT

The strains under review are ATCC-74040 and GHA. Strain ATCC-74040 was originally isolated from a boll weevil cadaver in Texas, USA. Strain GHA was originally isolated from the Southern corn rootworm, on green beans in Oregon, USA.

The representative formulated products for the evaluation are 'BotaniGard 22 WP' containing 4.4×10^{13} CFU/kg (strain GHA) and 'Naturalis-L' containing 2.3×10^{10} CFU/L (strain ATCC-74040).

The representative uses evaluated are on tomatoes for strain ATCC-74040 and on tomatoes, cucumbers and ornamentals for strain GHA. Full details of the GAP can be found in the lists of end points in Appendix A.

CONCLUSIONS OF THE EVALUATION

1. Identity of the microorganism/biological properties/physical and technical properties and methods of analysis.

Strain ATCC-74040 is maintained in the American Type Culture Collection. Strain GHA is also maintained in this culture collection under the accession number ATCC 74250.

The strains are not human pathogens and are not related to known human pathogens. The strains are not able to grow at or above 37 °C.

It is possible that these strains produce some toxins however insufficient information on this issue has been provided. The content of contaminating microorganisms was not fully addressed and a data gap has been identified. Validation of the biopotency method and other quantitative methods were not available. Also methods for the unequivocal identification to strain level were not available. Methods are not available for toxins and contaminating microorganisms.

For 'BotaniGard 22 WP' containing strain GHA the following data gaps were identified: storage stability, demonstrate that the poor wettability is not an issue, sprayability study to address suspensibility and wet sieve test.

For 'Naturalis-L' containing strain ATCC-74040 the following data gaps were identified: the proposed storage regime should be given in detail and then compared to the storage stability data.

2. Mammalian toxicity

General data

B. bassiana can produce a number of metabolites. For strain ATCC-74040, the only available information is about beauvericin suggesting that its concentration in the technical material is about 3 – 4 ppm. For the strain GHA it was shown that concentrations of approximately 50 ppm of beauvericin and bassianolide in a batch employed in the acute studies for oral, dermal, intratracheal and intraperitoneal toxicity did not exhibit adverse effects. Beauvericin, which was negative in an Ames test, was therefore proposed as an indicator; however, all the other metabolites were not defined. For both strains a data gap on strain specific data on the formation of toxins/secondary metabolites was identified (see section 1). Being a fungus, *Beauveria sp.* is not expected to possess the potential for transfer of genetic material.

Toxicity studies

Neither strain ATCC-74040 nor strain GHA showed any evidence of toxicity, infectivity or pathogenicity, based on acute oral (GHA LD₅₀ > 1 x 10⁸ CFU/animal, ATCC-74040 LD₅₀ > 1.9 x 10⁸ CFU/animal), intratracheal (GHA LD₅₀ > 1 x 10⁸ CFU/animal, ATCC-74040 LD₅₀ > 2.5 x 10⁹ CFU/animal) and intraperitoneal (GHA LD₅₀ > 1 x 10⁷ CFU/animal, ATCC-74040 LD₅₀ > 2.0 x 10⁷ CFU/animal) studies. An Ames test performed with both strains showed negative results.

However, in the intratracheal study with strain ATCC-74040 body weight decrease was noted in treated animals but not in the controls. Although the effect was clearly reversible, a data gap was identified for addressing the toxicity after repeated doses via inhalation (the details of the submitted published data on long-term toxicity do not allow for an independent assessment).

For both strains the warning phrase “Microorganisms may have the potential to provoke sensitising reactions”, can be applied taking into account that hazard statements applicable to chemicals (according to Regulation (EC) No 1272/2008¹⁰) are not appropriate for microorganisms.

Reference values

Based on the 1st tier studies, for strain GHA no reference values are needed as the microorganism was not shown to be pathogenic or infective based on the available data and studies. For strain ATCC-74040 it is considered unlikely that the microorganism is pathogenic or infective based on the 1st tier studies, not indicating the need of specific reference values; however, based on the submitted data, a data gap was identified to further clarify repeated inhalation toxicity.

Exposure estimates

Since reference values are not necessary, no exposure estimates are required for the microorganism. Due to the sensitisation potential of microorganisms observed in unprotected workers, the use of adequate personal protective equipment for operators and re-entry workers should be further considered for dermal and inhalatory exposure. Pending on the data gap for toxins/secondary metabolites the risk assessment for the operator, worker and bystander cannot be finalised. It is noted that bystander exposure is not relevant for the representative uses in greenhouse.

3. Residues

The microorganism is not pathogenic to humans. When applied to the plants it will only grow on/in the host organism and if toxins are produced they will only be contained inside the host, the dead host insect will either fall off or will be washed off. However, the consumer risk assessment cannot be finalised until the issue of toxins/secondary metabolites is addressed for section 1.

4. Environmental fate and behaviour

No information has been provided in relation to potential interference of *Beauveria bassiana* strains ATCC-74040 and GHA with the analytical systems for the control of the quality of drinking water provided for in Directive 98/83/EC¹¹ (see specific Annex VI decision making criteria in Directive 2005/25/EC¹²). However, as these methods require pathogenic bacteria to be identified and confirmed as absent, it is probably unlikely that filamentous fungi or their conidia would interfere with methodologies used for such determinations.

¹⁰ Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006. OJ L 353, 31.12.2008, p.1-1355.

¹¹ Council Directive 98/83/EC of 3 November 1998 on the quality of water intended for human consumption. OJ L 330, 5.12.98, p.32-54

¹² Council Directive 2005/25/EC of 14 March 2005 amending Annex VI to Directive 91/414/EEC as regards plant protection products containing micro-organisms. OJ L 90, 8.4.2005, p.1-34

No information has been provided on the potential transfer of genetic material from *Beauveria bassiana* strains ATCC-74040 and GHA to other organisms. Being a fungus, *Beauveria* sp. is not expected to possess plasmids in their cytoplasm (only mitochondrial plasmids are known). Consequently it is not expected to possess the potential for transfer of genetic material.

4.1. Fate and behaviour in the environment of the microorganism

Beauveria bassiana is naturally found throughout a wide range of habitats in temperate and tropical climates. The level of natural occurrence of *B. bassiana* strains ATCC-74040 and GHA in the soil is not precisely defined in the supplied literature. There are indications that *B. bassiana* strain GHA multiplies in the environment only in insect hosts, where it sporulates and disperses only under specific environmental conditions, particularly high humidity. In the absence of specific environmental conditions, the organism dies with its target host. However, a data gap has been identified for strain specific data on the environmental conditions required for dispersal and viability of conidia for *B. bassiana* strain GHA.

Three studies on the **persistence and multiplication in soil** of *B. bassiana* strain ATCC-74040 were submitted. However, uncertainties and deficiencies in terms of method of application, environmental conditions and origin of soils were identified for these studies. Overall, the available information is considered not sufficient to properly characterise the competitiveness/persistence of *B. bassiana* strains ATCC-74040 and GHA in soil in the context of the representative uses assessed. Therefore, a data gap has been identified. This data gap is also pertinent in respect of the representative use assessed for *B. bassiana* strains ATCC-74040 and GHA for greenhouse and indoor uses, in particular since what happens to the growth medium at the end of the production was not assessed in the dossier. This route of environmental exposure should be taken into account in territories where spent growing media can be spread on agricultural or horticultural land. The data gap is also relevant for production in temporary greenhouses (e.g. polythene tunnels) for the soil *in situ*.

A data gap was also identified for information on viability/population dynamics in soil-less media for the use of *B. bassiana* strains ATCC-74040 and GHA in greenhouse.

Little information has been provided on **persistence and multiplication in water** of *B. bassiana* strains ATCC-74040 and GHA. A study on the effect of pH and metal ions on *B. bassiana* strain GHA in aqueous suspension showed a loss in conidial viability in the different aqueous systems tested. In another study on the determination of storage stability of *B. bassiana* strain ATCC-74040 it was concluded that the microorganism is stable in water at pH 5, 7 and 9. No other information is given on persistence and multiplication in water. However, it is known from scientific publications that germination of conidia and therefore multiplication in water is not expected since *B. bassiana* is not an aquatic fungus and is therefore not adapted to the conditions of the aqueous environment.

Conidia of *B. bassiana* can be easily transported through the **air**. Also transmission of fungal spores through the air by insects is possible. On the basis of the information presented in the DAR and in Addendum 2 (March 2012) for *B. bassiana* strain GHA, it could be shown that viability of *B. bassiana* conidia in the air or epigeal habitats is mainly affected by the UV-B portion of the solar spectrum. On the basis of persistence data, it may be concluded that significant degradation of *B. bassiana* conidia on leaves under field conditions may occur within few days.

A study on **mobility in soil** showed that conidia of *B. bassiana* strain GHA are not very mobile in soil and generally remain on the soil surface. It was seen that the movement of conidia vertically through the soil profile was positively correlated with high infiltration rates in soil. Besides leaching, movement in the soil in horizontal and vertical direction is possible by soil arthropods, especially collembola. However, no groundwater risk assessment is necessary since *B. bassiana* is neither pathogenic nor toxic to humans.

4.2. Fate and behaviour in the environment of any relevant metabolite formed by the microorganism under relevant environmental conditions

A data gap has been identified for information on the production of toxins/secondary metabolites such as beauvericin (see section 1). If, under the conditions of use, relevant metabolites are present in or produced by the microorganism, data requirements and the corresponding risk assessment as outlined under Annex II, Part A point 7 of Directive 91/414/EC need to be fulfilled if all the following conditions are met:

- the relevant metabolite is stable outside the microorganism
- a toxic effect of the relevant metabolite is independent of the presence of the microorganism, and
- the relevant metabolite is expected to occur in the environment in concentrations considerably higher than under natural conditions.

This data gap is pertinent in respect of the representative uses in the field and greenhouse assessed for *B. bassiana* strain ATCC-74040 and for strain GHA for both indoor and greenhouse uses and following disposal of spent growing media, as what happens to the growth medium at the end of the production was not assessed. This route of environmental exposure would need to be taken into account in territories where spent growing media can be spread on agricultural or horticultural land.

5. Ecotoxicology

It is noted that a data gap was identified in relation to the formation of secondary metabolites/toxins, such as beauvericin, in the treated crops and in relevant environmental compartments after application of the product. Pending on the outcome of these data gaps, the potential risk of these secondary metabolites/toxins to non-target organisms has to be addressed (see Section 1 and Section 4.2).

Beauveria bassiana strain ATCC-74040

A strain-specific study on *Beauveria bassiana* strain ATCC-74040 was submitted to address the toxicity and pathogenicity to birds. The RMS noted a number of uncertainties in the study with regard to the possibility to detect infectivity and pathogenicity. However, the study was specifically designed to account for potential effects from infectivity and pathogenicity (5 days of exposure followed by 30 days observation period) and was considered to be useful for regulatory assessment. No indications of infectivity or pathogenicity were observed in the study and therefore a low risk to birds from infectivity and pathogenicity was concluded.

Beauveria bassiana strain ATCC-74040 was concluded not to be infectious or pathogenic to mammals (see Section 2); therefore a low risk to wild populations of mammals from infectivity and pathogenicity was concluded.

A study to address the toxicity, pathogenicity and infectiveness of *B. bassiana* strain ATCC-74040 on fish was available. A study on algae was available with the plant protection product 'Naturalis-L'. Overall, these studies were considered suitable to address the risk of infectiveness and pathogenicity, which was indicated as low. A study with *Daphnia magna* indicated an effect on immobilisation. The study was not considered sufficient for an assessment of infectivity and pathogenicity and therefore a data gap was concluded.

A strain specific study on bees for *B. bassiana* strain ATCC-74040 was provided. The available data submitted to address the effects on bees and pollinators were discussed at the Pesticides Peer Review Teleconference TC 67. No adverse effects were observed on honey bees, however since reproductive effects were observed for *Typhlodromus pyri*, experts agreed that the potential effects on bee brood are not covered by the available data for the representative field use. Furthermore, experts agreed that

further data are needed to address the reproductive effects of *B. bassiana* strain ATCC-74040 on pollinators for the representative field use. Therefore, data gaps were identified for these issues. This data gap is also relevant for pollinators which may be used in greenhouses, although it was acknowledged that bumble bees can be used to vector *B. bassiana* for insect control in greenhouses.

Some laboratory studies indicate that non-target arthropods may be susceptible to infection and pathogenicity from *B. bassiana* strain ATCC-74040. Information suggests that humid conditions will increase the likelihood of infection. With the information available, a high risk from infectivity and pathogenicity to non-target arthropods cannot be excluded.

An acute earthworm toxicity study performed with the product 'Naturalis-L' was available and indicated no adverse effects. It is acknowledged that the study was not specifically designed to investigate the potential for infectivity and pathogenicity; however, as the study was conducted for 14 days it is expected that there would have been some evidence of an effect. Therefore, on the basis of the available study a low risk to earthworms from infectivity and pathogenicity was concluded.

The risk to soil microorganisms was considered to be low for all representative uses.

No data were available to address the effects on biological methods for sewage treatment plants. A data gap was identified to address the risk for biological methods for sewage treatment plants for the greenhouse use.

Beauveria bassiana strain GHA

Some strain-specific studies to address the pathogenicity and infectiveness of *B. bassiana* strain GHA to fish and aquatic invertebrates were available. A study on algae with the plant protection product 'BotaniGard 22 WP' was also available. No indications of infectivity or pathogenicity were observed in the available fish study and therefore a low risk from infectivity and pathogenicity could be concluded. However, an adverse effect on the reproductive capacity was observed at the tested concentration and therefore no NOEC could be derived. The study was not considered sufficient to address the risk to fish from toxic effects and therefore further information is required (see data gap regarding the secondary metabolites and toxins, Section 1 and Section 4.2). The available data for aquatic invertebrates and algae indicated a low risk from infectivity and pathogenicity.

The risk assessment for pollinators which may be used in greenhouses was discussed at Pesticides Peer Review Teleconference TC 67. It was agreed that further information was required to demonstrate a low risk and therefore a data gap was identified.

A data gap was identified for information to complete the soil exposure assessment for *Beauveria bassiana* strain GHA, following disposal of spent growing media from greenhouses (see Section 4). Pending the outcome of the data gap it may be necessary to address the risk from infectivity and pathogenicity to soil-dwelling non-target arthropods and soil microorganisms. This is therefore identified as an assessment not finalised. A low risk to birds, mammals and earthworms from infectivity and pathogenicity was concluded.

No studies were available to address the effects on biological methods for sewage treatment plants and therefore a data gap was identified.

6. Overview of the risk assessment of compounds listed in residue definitions triggering assessment of effects data for the environmental compartments

6.1. Soil

Compound (name and/or code)	Persistence	Ecotoxicology
<i>Beauveria bassiana</i> strain ATCC-74040 and GHA	No data, data required	Low risk for strain <i>Beauveria bassiana</i> strain ATCC-74040. No data for soil microorganisms available for <i>Beauveria bassiana</i> strain GHA.
Relevant toxins or secondary metabolites	Data gap pending on their identification and quantification	Data gap pending on their identification and quantification

6.2. Ground water

Compound (name and/or code)	Mobility in soil	>0.1 µg/L 1m depth for the representative uses (at least one FOCUS scenario or relevant lysimeter)	Pesticidal activity	Toxicological relevance	Ecotoxicological activity
Relevant toxins or secondary metabolites	Data gap pending on their identification and quantification	Data gap pending on their identification and quantification	-	Data gap pending on their identification and quantification	Data gap pending on their identification and quantification

6.3. Surface water and sediment

Compound (name and/or code)	Ecotoxicology
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<i>Beauveria bassiana</i> strain ATCC-74040 and GHA	<i>Beauveria bassiana</i> strain ATCC-74040 – data gap to address the risk to aquatic invertebrates from infectivity and pathogenicity. <i>Beauveria bassiana</i> strain GHA – low risk from infectivity and pathogenicity.
Relevant toxins or secondary metabolites	No data available. Data gap pending on their identification and quantification.

6.4. Air

Compound (name and/or code)	Toxicology
<i>Beauveria bassiana</i> strain GHA	Not acutely toxic after intratracheal administration
<i>Beauveria bassiana</i> strain ATCC-74040	Not acutely toxic after intratracheal administration; data gap for repeated dose inhalation toxicity
Relevant toxins or secondary metabolites (both strains GHA and ATCC-74040)	No data available

7. List of studies to be generated, still ongoing or available but not peer reviewed

This is a complete list of the data gaps identified during the peer review process, including those areas where a study may have been made available during the peer review process but not considered for procedural reasons (without prejudice to the provisions of Article 7 of Directive 91/414/EEC concerning information on potentially harmful effects).

- Demonstrate that the level of microbial contamination complies with international standards. The OECD issues paper (OECD, 2011) should be used as the reference (relevant for strain ATCC 74040; submission date proposed by the notifier: studies available but not evaluated or peer reviewed; see section 1).
- Demonstrate that the level of microbial contamination complies with international standards. The OECD issue paper (OECD, 2011) should be used as the reference (relevant for strain GHA; submission date proposed by the notifier: unknown; see section 1).
- Validation of the biopotency method (relevant for all representative uses evaluated; submission date proposed by the notifiers: unknown; see section 1)
- Validation of the method to unequivocally identify the organism to strain level (relevant for all representative uses evaluated; submission date proposed by the notifiers: unknown; see section 1).
- Strain specific data on the formation of toxins/secondary metabolites. It is noted that some data are available for the compound beauvericin (relevant for strain ATCC 74040; submission date proposed by the notifier: unknown; see section 1 and 2).
- Strain specific data on the formation of toxins/secondary metabolites (relevant for strain GHA; submission date proposed by the notifier: unknown; see section 1 and 2).
- Validated quantitative methods for the microorganism in the technical material and formulated product (relevant for all representative uses evaluated; submission date proposed by the notifiers: unknown; see section 1).
- Validated methods for the microbial contaminants (relevant for strain ATCC 74040; submission date proposed by the notifier: studies available but not evaluated or peer reviewed; see section 1).
- Validated methods for the microbial contaminants (relevant for strain GHA; submission date proposed by the notifier: unknown; see section 1).
- Validated method of analysis for toxins (relevant for all representative uses of strain GHA evaluated; submission date proposed by the notifier: unknown; see section 1).
- The proposed storage conditions for the formulated product should be given in detail and it should be checked if the storage stability data support the proposed conditions (relevant for all representative uses of 'Naturalis-L' evaluated; submission date proposed by the notifier: unknown; see section 1).
- Storage stability data; demonstration that during normal use the poor wettability does not cause a problem; sprayability study to demonstrate that during normal use the poor suspensibility is not an issue; wet sieve test (relevant for all representative uses of 'BotaniGard 22 WP' evaluated; submission date proposed by the notifier: unknown; see section 1).
- Further clarification on the repeated inhalation toxicity of ATCC-74040 (relevant for all representative uses of strain ATCC-74040; submission date proposed by the notifier: unknown; see section 2).

- Satisfactory information to demonstrate that *Beauveria bassiana* strains ATCC-74040 and GHA will not persist in soil in concentrations higher than the natural background levels, taking into account repeated applications over the years. Pending the outcome of the data gap, a risk assessment for soil-dwelling non-target arthropods and soil microorganisms may be required (relevant for all representative uses evaluated and in territories where spent growing media from indoor/greenhouse uses can be spread on agricultural or horticultural land or in temporary greenhouses (e.g. polythene tunnels) for the soil *in situ*; submission data proposed by the notifiers: unknown; see section 4 and 5).
- Satisfactory information to demonstrate that, under the conditions of use, any relevant toxins/secondary metabolites produced by *Beauveria bassiana* strains ATCC-74040 and GHA (which may include beauvericin) will not occur in the environmental compartments in concentrations considerably higher than under natural conditions. Further data on the persistence, transformation and mobility of these compounds may be needed in order to assess the potential level of environmental exposure including groundwater (relevant for all representative uses evaluated; submission date proposed by the notifiers: unknown; see section 4).
- Strain specific data on the environmental conditions required for dispersal and viability of conidia for *B. bassiana* strain GHA in the environment (relevant for all representative uses of strain GHA; submission date proposed by the notifier: unknown; see section 4).
- Information on viability/population dynamics in soil-less media for the use of *B. bassiana* strains ATCC-74040 and GHA in greenhouse (relevant for greenhouse uses; submission date proposed by the notifiers: unknown; see section 4).
- Pending on the outcome of the data gaps identified for the secondary metabolites and toxins, the risk to non-target organisms may need to be addressed (relevant for representative uses of *B. bassiana* strains ATCC-74040 and GHA; submission date proposed by the notifier: unknown; see section 5).
- Further data to address the potential for infectivity and pathogenicity to aquatic invertebrates (relevant for all representative uses of *B. bassiana* strain ATCC-74040; submission date proposed by the notifier: unknown; see section 5).
- Further data to address the potential effect on honey bee brood (relevant for representative field use of *B. bassiana* strain ATCC-74040; submission date proposed by the notifier: unknown; see section 5).
- Further data to address the reproductive effect of *B. bassiana* strains ATCC-74040 and GHA to pollinators (relevant for all representative uses of *B. bassiana* strains ATCC-74040 and GHA; submission date proposed by the notifier: unknown; see section 5).
- Further data to address the potential for infectivity and pathogenicity to non-target arthropods (relevant for field representative uses of *B. bassiana* strain ATCC-74040; submission date proposed by the notifier: unknown; see section 5).
- Information to address the effects on biological methods for sewage treatment plants for the use of *B. bassiana* strains ATCC-74040 and GHA in greenhouse (relevant for greenhouse uses; submission date proposed by the notifier: unknown; see section 5).

8. Particular conditions proposed to be taken into account to manage the risk(s) identified

- none

9. Concerns

9.1. Issues that could not be finalised

An issue is listed as an issue that could not be finalised where there is not enough information available to perform an assessment, even at the lowest tier level, for the representative uses in line with the Uniform Principles of Annex VI to Directive 91/414/EEC and where the issue is of such importance that it could, when finalised, become a concern (which would also be listed as a critical area of concern if it is of relevance to all representative uses).

1. The production of relevant toxins/secondary metabolites cannot be excluded and therefore the risk assessment cannot be finalised for humans (including operators, workers, bystanders and consumers) and the environment including the assessment of potential groundwater contamination and the potential risk to non-target organisms.
2. Satisfactory information is not available to demonstrate that *B. bassiana* strains ATCC-74040 and GHA will not persist in soil in concentrations higher than the natural background levels, taking into account repeated applications over the years. Consequently, the level of exposure in soil of *B. bassiana* strain ATCC-74040 cannot be reliably estimated for the field uses. This applies also to the use of *B. bassiana* strains ATCC-74040 and GHA in temporary greenhouse systems (e.g. polythene tunnels) or when spent growing media is spread on agricultural or horticultural land. Pending the outcome of the data gap a risk assessment for soil-dwelling non-target arthropods and soil microorganisms from *B. bassiana* strain GHA may be required.
3. The risk assessment for bees and other pollinators cannot be finalised with the available data for the representative uses of *B. bassiana* strain ATCC-74040 and strain GHA.

9.2. Critical areas of concern

An issue is listed as a critical area of concern where there is enough information available to perform an assessment for the representative uses in line with the Uniform Principles of Annex VI to Directive 91/414/EEC, and where this assessment does not permit to conclude that for at least one of the representative uses it may be expected that a plant protection product containing the active substance will not have any harmful effect on human or animal health or on groundwater or any unacceptable influence on the environment.

An issue is also listed as a critical area of concern where the assessment at a higher tier level could not be finalised due to a lack of information, and where the assessment performed at the lower tier level does not permit to conclude that for at least one of the representative uses it may be expected that a plant protection product containing the active substance will not have any harmful effect on human or animal health or on groundwater or any unacceptable influence on the environment.

- none

9.3. Overview of the concerns identified for each representative use considered

(If a particular condition proposed to be taken into account to manage an identified risk, as listed in section 8, has been evaluated as being effective, then 'risk identified' is not indicated in this table.)

Representative use		Tomatoes Greenhouse ATCC-74040	Tomatoes Field ATCC-74040	Ornamentals Indoor GHA	Tomatoes, cucumber Greenhouse GHA
Operator risk	Risk identified				
	Assessment not finalised	X ¹	X ¹	X ¹	X ¹
Worker risk	Risk identified				
	Assessment not finalised	X ¹	X ¹	X ¹	X ¹
Bystander risk	Risk identified				
	Assessment not finalised		X ¹		
Consumer risk	Risk identified				
	Assessment not finalised	X ¹	X ¹		X ¹
Risk to wild non target terrestrial vertebrates	Risk identified				
	Assessment not finalised	X ¹	X ¹	X ¹	X ¹
Risk to wild non target terrestrial organisms other than vertebrates	Risk identified		X		
	Assessment not finalised	X ^{1, 3}	X ^{1, 3}	X ^{1, 2, 3}	X ^{1, 2, 3}
Risk to aquatic organisms	Risk identified				
	Assessment not finalised	X ¹	X ¹	X ¹	X ¹
Groundwater exposure active substance	Legal parametric value breached				
	Assessment not finalised				
Groundwater exposure metabolites	Legal parametric value breached				
	Parametric value of 10µg/L ^(a) breached				
	Assessment not finalised	X ¹	X ¹	X ¹	X ¹
Comments/Remarks					

The superscript numbers in this table relate to the numbered points indicated in sections 9.1 and 9.2. Where there is no superscript number see sections 2 to 6 for further information.

(a): Value for non-relevant metabolites prescribed in SANCO/221/2000-rev 10-final, European Commission, 2003

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- EFSA (European Food Safety Authority), 2012. Peer Review Report to the conclusion regarding the peer review of the pesticide risk assessment of the active substance *Beauveria bassiana* strains ATCC-74040 and GHA.
- European Commission, 2003. Guidance Document on Assessment of the Relevance of Metabolites in Groundwater of Substances Regulated under Council Directive 91/414/EEC. SANCO/221/2000-rev. 10 - final, 25 February 2003.
- European Commission, 2008a. Review Report for the active substance *Beauveria bassiana* strain ATCC-74040 finalised in the Standing Committee on the Food Chain and Animal Health at its meeting on 11 July 2008 in view of the inclusion of *Beauveria bassiana* strain ATCC-74040 in Annex I of Directive 91/414/EEC. SANCO/1546/08 – rev. 3, 07 May 2008.
- European Commission, 2008b. Review Report for the active substance *Beauveria bassiana* strain GHA finalised in the Standing Committee on the Food Chain and Animal Health at its meeting on 11 July 2008 in view of the inclusion of *Beauveria bassiana* strain GHA in Annex I of Directive 91/414/EEC. SANCO/1547/08 – rev. 3, 07 May 2008.
- FOCUS, 2008. “Pesticides in Air: Considerations for Exposure Assessment”. Report of the FOCUS Working Group on Pesticides in Air, EC Document Reference SANCO/10553/2006 Rev 2 June 2008.
- Germany, 2007a. Draft Assessment Report (DAR) on the active substance *Beauveria bassiana* strain ATCC-74040 prepared by the rapporteur Member State Germany in the framework of Directive 91/414/EEC, November 2007.
- Germany, 2007b. Draft Assessment Report (DAR) on the active substance *Beauveria bassiana* strain GHA prepared by the rapporteur Member State Germany in the framework of Directive 91/414/EEC, November 2007.
- Germany, 2012a. Final Addendum to Draft Assessment Report on *Beauveria bassiana* strain ATCC-74040, compiled by EFSA, May 2012.
- Germany, 2012b. Final Addendum to Draft Assessment Report on *Beauveria bassiana* strain GHA, compiled by EFSA, May 2012.
- OECD (Organisation for Economic Co-operation and Development), 2011. OECD Issue Paper on Microbial Contaminant Limits for Microbial Pest Control Products, ENV/JM/MONO(2011)43.

APPENDICES

Appendix A – list of end points for the active substance *Beauveria bassiana* strain ATCC-74040 and the representative formulation

Chapter 1: Identity, Biological properties, Details of uses, Further information, and Proposed Classification and Labelling

Active micro-organism:	<i>Beauveria bassiana</i> (Balsamo) Vuillemin
Function (e.g. control of fungi):	Control of insects
Rapporteur Member State	Germany

Identity of the Microbial Pest control Agent / Active substance (OECD data point IIM 1)

Name of the organism:	<i>Beauveria bassiana</i> (Balsamo) Vuillemin
Taxonomy:	Phylum: Deuteromycota, Class: Hyphomycetes, Order: Moniliales, Family: Moniliaceae, Genus: Beauveria
Species, subspecies, strain:	<i>B. bassiana</i> strain ATCC 74040
Identification / detection:	The organism has been identified according to the taxonomic descriptions of de Hoog 1972 (in particular conidiospore morphology) analysed by standard laboratory microscopy. Open for specific method.
Culture collection:	<i>B. bassiana</i> strain ATCC 74040 is maintained in the American Culture Collection under ATCC 74040.
Minimum and maximum concentration of the MPCA used for manufacturing of the formulated product (viable granules; g/kg):	2×10^{10} CFU/g in the technical MPCA with a minimum of 1.6×10^{10} CFU/g.
Identity and content of relevant impurities, additives, contaminating organisms in the technical grade of MPCA:	none
Is the MCPA genetically modified; if so provide type of modification	Not applicable

Biological properties of the micro-organism (OECD data point IIM 2)

Origin and natural occurrence, Background level:	<i>Beauveria bassiana</i> strain ATCC 74040 is an entomopathogenic fungus which is a common pathogen of many insect species. It is a ubiquitous fungus, found in a variety of soils from a broad range of climatic conditions in most countries.
Target organism(s):	Whiteflies (<i>Aleurodidade</i>)
Mode of action:	The conidia of <i>B. bassiana</i> strain ATCC 74040 adhere to the insect cuticle by means of hydrophobic interaction between the spore wall and epicuticle lipids. The conidia germinate,

	and the germ tube penetrates the cuticle, using a specific series of enzymes, which in turn degrade the lipids, protein and chitin in the insect cuticle. In the insect body, the fungus multiplies in the haemocoel as a blastospore, or yeast-like cell, and enzymes begin to destroy the internal structures of the host insect causing morbidity within 36 - 72 hours. Reduced feeding and immobility are rapidly evident, and the insect dies within 4 to 10 days post-infection. The time to death will depend on the insect species, age and conidial dose. After death, the blastospores transform into mycelia, which emerge through the cuticle and form spores. These cover the cadaver as a characteristic white mould. Sporulation occurs only in conditions of high humidity.
Host specificity:	Infection mechanisms are specific to insects. Not pathogenic to humans / plants.
Life cycle:	The conidia adhere to the insect cuticle, germinate and penetrate in the insect body, where they replicate as yeast-like cells (blastospores) and destroy the internal structures, causing morbidity within 36 - 72 hours. After death of the insect, the blastospores transform into mycelia, which emerge through the cuticle and form spores. <i>B. bassiana</i> strain ATCC 74040 has no resting stages and no known sexual stage.
Infectivity, dispersal and colonisation ability:	In general, <i>B. bassiana</i> grows in a wide temperature range from 5 to 35 °C. The optimal growth temperature for <i>B. bassiana</i> is 23 to 28°C. Maximum temperature for the majority of analysed strains is below 35°C. Water activity below 0.99 already leads to considerable reductions in growth and no growth was observed at 0.88. <i>B. bassiana</i> strain ATCC 74040 remained viable in aqueous suspensions at pH 5, 7, and 9 for 24 hours in a laboratory test system. The germination rate of conidia decreased rapidly after exposition to UV light, and short wave UV was more effective in inactivation than long-wave UV. No viable spores were detected after 32 min. of long-wave-UV irradiation.
Relationships to known plant, animal or human pathogens:	<i>Beauveria bassiana</i> strain ATCC 74040 is not related to any known human or plant pathogen.
Genetic stability:	During more than 10 years of commercial use and production, no change has occurred in phenotypic characteristics or RAPD analyses of <i>Beauveria bassiana</i> Strain ATCC 74040.
Information on the production of relevant metabolites (especially toxins):	Beauvericin: max content 5 mg/kg Open for further studies.
Resistance/ sensitivity to antibiotics / anti-microbial agents used in human or veterinary medicine:	No specific information addressing this issue has been submitted by the notifier or could be found in the open literature. However, because spread of resistance (genes) to antimycotic agents between fungal species is not of that

concern as for bacteria with regard to antibiotics, and since *Beauveria bassiana* itself is only an opportunistic pathogen that has very rarely caused infections in humans and is not related to any known human pathogen, this lack of information is not considered a critical data gap.

Summary of uses supported by available data (*Beauveria bassiana* strain ATCC 74040)

Crop and/or situation (a)	Member State or Country	Product name	F,G or I (b)	Pests or Group of pests controlled (c)	Formulation		Application				Application rate per treatment MPCA			PHI (days) (m)	Remarks:
					Type (d-f)	Conc. of as (i)	Method kind (f-h)	Growth stage & season (j)	Number min max (k)	Interval between applications	g as/hL min max	water L/ha min max	g as/ha min max		
Tomatoes	Europe	Naturalis-L	G	Whiteflies	OD	0.0691 kg/L (2.3 x 10 ¹⁰ CFU/L)	Spray	-	3 - 5	5 - 7 d	5.18 - 6.91 (1.7 x 10 ⁹ - 2.3 x 10 ⁹ CFU/hL)	(800) - 1000	51.8 - 69.1 (1.7 x 10 ¹⁰ - 2.3 x 10 ¹⁰ CFU/ha)	NA	Rate of MPCA: 0.75-1 L/ha
Tomatoes	Europe	Naturalis-L	F	Whiteflies	OD	0.0691 kg/L (2.3 x 10 ¹⁰ CFU/L)	Spray	-	3 - 5	5 - 7 d	5.18 - 6.91 (1.7 x 10 ⁹ - 2.3 x 10 ⁹ CFU/hL)	(800) - 1000	51.8 - 69.1 (1.7 x 10 ¹⁰ - 2.3 x 10 ¹⁰ CFU/ha)	NA	Rate of MPCA: 0.75-1 L/ha

- (a) For crops, the EU and codex classifications (both) should be used; where relevant, the situation should be described (e.g. fumigation of a structure)
- (b) Outdoor or field use (F), glasshouse application (G) or indoor application (I)
- (c) e.g. biting and sucking insects, soil born insects, foliar fungi, weeds
- (d) e.g. wettable powder (WP), emusifiable concentrate (EC), granule (GR)
- (e) GCPF codes - Crop Life Technical Monograph No 2, 1989
- (f) All abbreviations used must be explained
- (g) Method, e.g. high volume spraying, low volume spraying, spreading, dusting, drench
- (h) Kind, e.g. overall, broadcast, aerial spraying, row, individual plant, between the plant - type of equipment used must be indicated
- (i) CFU/kg or CFU/L
- (j) Growth stage at last treatment (BBCH Monograph, growth stages of Plants, 1997, Blackwell ISBN 3-8263-3152-4), including where relevant, information on season at time of application
- (k) Indicate the minimum and maximum number of application possible under practical conditions of use
- (l) The volume should be given in g or kg whatever gives the more readable number (eg. 200 kg/ha instead of 200000g/ha) as well as in number of CFU
- (m) PHI - minimum pre-harvest interval

Classification and proposed labelling (Symbol, Indication of danger, Risk phrases, Safety phrases)

with regard to physical/chemical data:	none
with regard to toxicological data	R42 – May cause sensitisation by inhalation R43 – May cause sensitisation by skin contact
with regard to fate and behaviour	Not applicable
with regard to ecotoxicological data	Not applicable

Chapter 2: Methods of analysis

Analytical methods for the micro-organism (OECD data point IIM 4.2, 4.3 and IIM 5.3)

manufactured micro-organism (principal of method):	Analytical methods for the identification of the technical <i>Beauveria bassiana</i> strain ATCC 74040 include the following: - Characterisation by light microscopic examination of the visual morphology - Random Amplified Polymorphic DNA (RAPD) analysis of fractionated total genomic DNA. - Discrimination of the strain from other strains of the same species by microsatellites. - Determination of fungal spore-count and spore viability by haemocytometer counts and <i>in vitro</i> germination assay. - Open for validation
Impurities and contaminating micro-organisms in manufactured material (principal of method):	Enumeration of microbial contaminants, determining the presence/absence of enteric bacteria such as <i>Shigella</i>, <i>Salmonella</i> and <i>Vibrio spp.</i> by plating out on selective media (Thiosulfate-Citrate-Bile-Sucrose and Salmonella-Shigella agars). The possible presence of impurities such as Aflatoxin and Beauvericin is assessed by HPLC. HPLC-UV (column switch) for beauvericin Open for contaminating microorganisms
Microbial Pest control product (principle of method):	Analytical methods for the formulation are as used for the active substance.

Analytical methods for residues (viable and non-viable) in exposed compartments and organisms (OECD data point IIM 4.5)

of the active micro-organism (principle of method)	Validated methods are not required. However, for water a method is available and validated with an LOQ of 6×10^7 CFU/L
Of relevant metabolites (principle of method):	none relevant

Chapter 3: Impact on Human and Animal Health

Medical data: (including medical surveillance on manufacturing plant personnel) (OECD data point IIM 5.1, 5.2)	No adverse effects in employees who had been in occupational contact with strain ATCC reported. For <i>Beauveria bassiana</i> in general, few cases of local (eye) and systemic infections were reported with the latter occurring only in immunocompromised or severely ill patients. Evidence of allergenicity by respiratory or skin contact.
Sensitisation: (OECD data point IIM 5.3.1 & IIM 7.1.6)	Microorganisms may have the potential to provoke sensitising reactions For <i>Beauveria bassiana</i> in general, there is evidence of allergenicity by inhalation in laboratory animals and humans and indications of allergic skin reactions in humans. The available supplementary M&K test with strain ATCC in guinea pigs, although negative, was not sufficient to contravene these findings.
Acute oral infectivity, toxicity and pathogenicity (OECD data point IIM 5.3.2 & IIM 7.1.1)	Rat: LD ₅₀ > 1.9 x 10 ⁸ CFU/animal; no evidence of toxicity, pathogenicity, or infectivity. Spores still detectable in faeces at end of study on day 14.
Acute intratracheal/inhalation toxicity, pathogenicity and infectivity: (OECD data point IIM 5.3.3 & IIM 7.1.3)	Rat: LD ₅₀ > 2.5 x 10 ⁹ CFU/animal; initial bw decrease, local reversible signs of lung inflammation. No evidence of pathogenicity or infectivity. Clearance in lung tissue completed by day 15 post-dosing.
Acute intravenous/intraperitoneal toxicity, pathogenicity and infectivity: (OECD data point IIM 5.3.4 & IIM 7.1.2)	Rat: LD ₅₀ > 2.0 x 10 ⁷ CFU; no evidence of toxicity (nasal secretion on day 1), pathogenicity or infectivity. Clearance in blood completed by day 2 post-dosing.
Genotoxicity: (OECD data point IIM 5.3.5)	Negative in the Ames test
Cell culture study: (OECD data point IIM 5.3.6)	No data – not required.
Information on short-term toxicity and pathogenicity: (OECD data point IIM 5.3.7)	No data – not required.
Dermal toxicity: (OECD data point IIM 7.1.2)	Rabbit: LD ₅₀ > 1 x 10 ¹⁰ CFU/animal; no systemic effects, slightly irritating to the skin (no classification needed).
Specific-toxicity, pathogenicity and infectivity: (OECD data point IIM 5.5)	Rabbit: LD ₅₀ > 1 x 10 ¹⁰ CFU/animal; no systemic effects, slightly irritating to the skin (no classification needed). Certain potential of the microorganism (strain not specified) to damage the cornea confirmed in rabbits. Long-term inhalative exposure to <i>B. bassiana</i> (strain not specified) caused allergic lung reactions in rats and mice. Data gap for specific repeated dose toxicity
Genotoxicity – <i>in vivo</i> studies in germ cells: (OECD data point IIM 5.5.3)	No data – not required.

Additional studies with metabolites

Toxicological studies with strain GHA including a beauvericin and bassianolide contamination with approx. 50 mg/kg each indicated no additional adverse effects after oral, dermal or ip administration. Effects seen after intratracheal administration are unlikely to be due to beauvericin and bassianolide.

Beauvericin was not mutagenic in an Ames test.

Exposure to (the sum of) the metabolites should stay below 1.5 µg/d (threshold of toxicological concern).

Reference values

AOEL

Not established, not needed¹

ARfD

Not established, not needed¹

ADI

Not established, not needed¹

¹ based on the submitted data, a data gap was identified to further clarify repeated inhalation toxicity

Exposure (operator, workers, bystander, consumer):

(OECD data point IIM 6.1, IIM 7.2, 7.3 and 8.0)

Operator

Not finalised (lack of information on toxin/metabolites production)

Bystander

Not finalised (lack of information on toxin/metabolites production)

(For greenhouse applications, the bystander exposure is not relevant).

Worker

Not finalised (lack of information on toxin/metabolites production)

Chapter 4: Residues

Open for toxins/secondary metabolites

Chapter 5: Fate and Behaviour in the Environment (OECD IIM 7 & IIM 9)

Persistence and multiplication

in soil:

Data gap

The available data on measurement or observation of persistence and multiplication in soil was insufficient.

in water:

In an aquatic environment spores will be subject to sedimentation and can be expected to degrade rapidly in natural water bodies.

Based on read across from strain GHA:

The conidia did not germinate in different tested aqueous systems (distilled water, and aqueous

in air:

suspensions at pH 5, 7 and 9, or in metal ion solutions in the presence of bacteria).

Based on read across from strain GHA:

In the absence of a specific host insect, viability of conidia of *B. bassiana* in the air or epigeal habitats is mainly affected by the UV-B portion of the solar spectrum. On the basis of the persistence data, significant degradation of *B. bassiana* conidia on leaves under field conditions may occur in less than 2 days.

Mobility:

In soil:

Based on read across from strain GHA:

Conidia of *B. bassiana* are not very mobile in soil and generally remain on the surface of the soil. No groundwater risk assessment is necessary since *B. bassiana* is neither pathogenic nor toxic to humans.

In water: The globose or ovoid shape of conidia is not suitable for efficient transport in water.

In air:

Beauveria bassiana is dry, of small size and it is produced in powdery clusters. Therefore, these types of conidia can be easily transported through the air. Also transmission of fungal spores through the air by insects is possible.

However, in the absence of specific host insects, conidia of *Beauveria bassiana* will not persist in the sun-exposed phylloplane in significant amount for more than 2 days.

Chapter 6: Effects on non-target organisms (OECD IIM 8 & IIM 10)

Effects on birds (OECD IIM 8.1 & IIM 10.1)

Application rate (kg MPCA/ha)	Crop	Species	Time-scale	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
0.0691 (corresponding to 2.3×10^{10} CFU/ha)	Tomato, N/S-Europe (field)	Bobwhite quail	5 days dosing / 25 days observation	LD ₅₀ > 2667 mg/kg bw/day corresponding to 6.85×10^{10} CFU/kg bw/day (No observed external signs of pathogenicity. No findings of the gross necropsy)

Effects on mammals

Application rate (kg MPCA/ha)	Crop	Species	Time-scale	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
0.0691 (corresponding to 2.3×10^{10} CFU/ha)	Tomato, N/S-Europe (field)	Rat	Acute	Rat: LD ₅₀ > 1.9×10^8 CFU/animal; no evidence of toxicity, pathogenicity, or infectivity. Spores still detectable in faeces at end of study on day 14.

Effects on aquatic organisms (OECD IIM 8.2, 8.3 & IIM 10.2)

Group	Test substance	Time-scale	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
Laboratory tests			
Fish species: <i>Pimephales promelas</i>	<i>B. bassiana</i> ATCC 74040	30 days	NOEC < 1×10^9 CFU/L (survival of 26 days post hatch; larval growth)
Invertebrate species: <i>Daphnia magna</i>	<i>B. bassiana</i> ATCC 74040	21 days	Effects at lowest concentration tested. (Mortality of adults)

Effects on algae: (growth, growth rate, capacity to recover) (OECD IIM 8.4 & IIM 10.2)	<i>Pseudokirchneriella subcapitata</i> ; Naturalis-L; 3 d; E _r C ₅₀ : > 100 mg/L; E _b C ₅₀ : > 100 mg/L; NOE _b C: < 100 mg/L; NOE _r C: > = 100 mg/L 100 mg corresponds to 2.3×10^6 CFU/L or 7.16 mg as/L
Effects on aquatic plants (OECD IIM 8.5 & IIM 10.2)	Not performed, not required

Effects on bees (OECD IIM 8.7 & IIIM 10.3)

Application rate (kg MPCA/ha)	Crop	Route	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
Laboratory Tests			
	Tomato, N/S-Europe (field)	one time contact, 30-day continuous dietary exposure	Averaged survival rates: 92 % in the untreated control 78 % in the attenuated control 78 % among treated honeybees
Laboratory Tests (bumblebees)			
approx. 2.88×10^7 CFU/L	Tomato, N/S-Europe (field)	dermal contact, orally via sugar water, orally via pollen	$LC_{50} > 3.45 \times 10^7$ CFU/L

Effects on terrestrial arthropods other than bees (OECD IIM 8.8 & IIIM 10.4)

Species	Stage	Test Substance	Dose kg MPCA/ha)	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
Laboratory Tests				
<i>Aphidius rhopalosiphi</i>	adult	Naturalis-L	5.6, 11.6, 24, 50 and 104 g a.s./ha (0.081, 0.168, 0.347, 0.724 and 1.505 L product/ha)	LR ₅₀ : 68.9 g MPCA/ha (0.997 L Naturalis-L/ha) Corresponding to 2.3×10^{10} CFU/ha NOEC: 24 g MPCA/ha (0.347 L Naturalis-L/ha)
<i>Typhlodromus pyri</i>	adult	Naturalis-L	50, 104, 216, 450 and 934 g a.s./ha, equivalent to 0.723, 1.505, 3.125, 6.512 and 13.515 L Naturalis-L/ha	LR ₅₀ : 402.1 g a.s./ha, equivalent to 5.819 L Naturalis-L/ha (Mortality) Corresponding to 1.34×10^{11} CFU/ha ER ₅₀ : 81 g a.s./ha; equivalent to 1.1 L Naturalis-L/ha (Fecundity) Corresponding to 2.53×10^{10} CFU/ha NOEC: 50 g MPCA/ha (0.723 L Naturalis-L/ha)
Extended Laboratory Tests				
<i>Aphidius rhopalosiphi</i>	adult	Naturalis-L	50, 104, 221, 470 and 1000 g a.s./ha, corresponding to 0.698,	LR ₅₀ : > 13.66 L/ha (> 1000 g MPCA/ha) (Mortality); ER ₅₀ : > 13.66 L/ha (> 1000 g MPCA/ha) (Reproduction) Corresponding to $> 3.14 \times 10^{11}$ CFU/ha

			1.453, 3.087, 6.564 and 13.966 L product/ha	NOEC: 13,66 L/ha (> 1000 g MPCA/ha)
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Effects on other terrestrial invertebrates (OECD IIM 8.9, 8.9.1, 8.9.2 & IIM 10.5)

Toxicity, infectivity and pathogenicity: (endpoint, value or other description of effects)	LC ₅₀ (Mortality): > 1000 mg Naturalis-L/kg artificial soil > 2.3 x 10 ⁷ CFU/kg artificial soil (<i>Eisenia foetida</i>)
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Effects on soil microorganisms (OECD IIM 8.10 & IIM 10.6)

1.29 and 12.87 mg Naturalis-L/kg soil dry weight, corresponding to 1 and 10 L Naturalis-L/ha or 0.092 and 0.92 mg as/kg or 2.98 x 10 ⁴ and 29.8 x 10 ⁴ CFU/kg, silty sand soil (pH 6.70) <u>short-term respiration:</u> Deviation from control after 28 days < 25% in both concentrations <u>nitrogen turnover:</u> Deviation from control in the lower concentration after 28 days: < 25% Deviation from control at highest concentration after 28 days: > 25% Deviation from control at highest concentration after 56 days: < 25

Additional studies (OECD IIM 8.11 & IIM 10.7)

No data available for the biological methods of sewage treatment plants. Data gap was identified for the biological methods of sewage treatment plants for the greenhouse use.
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Appendix B – list of end points for the active substance *Beauveria bassiana* strain GHA and the representative formulation

Chapter 1: Identity, Biological properties, Details of uses, Further information, and Proposed Classification and Labelling

Active micro-organism:	<i>Beauveria bassiana</i> (Balsamo) Vuillemin
Function (e.g. control of fungi):	Control of insects
Rapporteur Member State	Germany

Identity of the Microbial Pest control Agent / Active substance (OECD data point IIM 1)

Name of the organism:	<i>Beauveria bassiana</i> (Balsamo) Vuillemin
Taxonomy:	Phylum: Deuteromycota, Class: Hyphomycetes, Order: Moniliales, Family: Moniliaceae, Genus: Beauveria
Species, subspecies, strain:	<i>B. bassiana</i> strain GHA
Identification / detection:	The organism has been identified according to the taxonomic descriptions of de Hoog 1972 (in particular conidiospore morphology) analysed by standard laboratory microscopy. Open for specific method.
Culture collection:	<i>B. bassiana</i> strain GHA is maintained in the American Culture Collection under ATCC 74250.
Minimum and maximum concentration of the MPCA used for manufacturing of the formulated product (viable granules; g/kg):	Nominal (mean) purity: 1.4×10^{11} viable CFU/g in the technical material. Acceptable range: 1.24×10^{11} to 1.47×10^{11} viable CFU/g
Identity and content of relevant impurities, additives, contaminating organisms in the technical grade of MPCA:	none
Is the MCPA genetically modified; if so provide type of modification	Not applicable

Biological properties of the micro-organism (OECD data point IIM 2)

Origin and natural occurrence, Background level:	<i>Beauveria bassiana</i> strain GHA is an entomopathogenic fungus which is a common pathogen of many insect species. It is a ubiquitous and cosmopolitan fungus, found in a variety of soils from a broad range of climatic conditions in most countries.
Target organism(s):	Whiteflies (<i>Aleurodidae</i>), thrips (<i>Thysanoptera</i>) and aphids (<i>Aphididae</i>).
Mode of action:	The conidia of <i>B. bassiana</i> strain GHA adhere to the insect cuticle by means of hydrophobic interaction between the spore wall and epicuticle lipids. The conidia germinate, and the germ tube penetrates the cuticle, using a specific series of enzymes, which in turn degrade the lipids, protein and chitin in the insect cuticle. In the insect body, the fungus multiplies

	in the haemocoel as a blastospore, or yeast-like cell, and enzymes begin to destroy the internal structures of the host insect causing morbidity within 36 - 72 hours. Reduced feeding and immobility are rapidly evident, and the insect dies within between 4 to 10 days post-infection. The time to death will depend on the insect species, age and conidial dose. After death, the blastospores transform into mycelia, which emerge through the cuticle and form spores. These cover the cadaver as a characteristic white growth. Sporulation occurs only in conditions of high humidity.
Host specificity:	<i>B. bassiana</i> is specific to insects. Not pathogenic to humans or plants.
Life cycle:	The conidia adhere to the insect cuticle, germinate and penetrate in the insect body, where they replicate as yeast-like cells (blastospores) and destroy the internal structures, causing morbidity within 36 - 72 hours. After death of the insect, the blastospores transform into mycelia, which emerge through the cuticle and form spores.
Infectivity, dispersal and colonisation ability:	<i>B. bassiana</i> strain GHA has an optimal growth temperature range between 23°C and 28°C. The conidia are not able to germinate or grow at $\geq 36^\circ\text{C}$. <i>B. bassiana</i> strain GHA is stable in water at pH 5, 7 and 9 and has been shown to be unaffected when exposed to metallic ions (Na, Ca, Fe, Mg). Detectable effects have only been noted when the fungus is exposed to copper, which is a known fungicide. Under natural conditions, the conidia of <i>B. bassiana</i> germinate and die in the absence of a suitable host insect within two days, and in aqueous environments. The conidia survive naturally in sheltered habitats and require specific environmental conditions of moderate temperature, high humidity and high insect population density for epizootic spread and dispersal.
Relationships to known plant, animal or human pathogens:	<i>Beauveria bassiana</i> strain GHA is not related to any known human or plant pathogen.
Genetic stability:	During more than 10 years of commercial use and production, no change has occurred in phenotypic characteristics or RFLP cDNA fingerprint of <i>B. bassiana</i> strain GHA.
Information on the production of relevant metabolites (especially toxins):	Beauvericin: max 5 mg/kg Open for further studies.
Resistance/ sensitivity to antibiotics / anti-microbial agents used in human or veterinary medicine:	No specific information addressing this issue has been submitted by the notifier or could be found in the open literature. However, because spread of resistance (genes) to antimycotic agents between fungal species is not of that concern as for bacteria with regard to antibiotics, and since <i>Beauveria bassiana</i> itself is only an opportunistic pathogen that has very rarely caused infections in humans and is not related to any known human pathogen, this lack of information is not considered a critical data gap.

Summary of uses supported by available data (*Beauveria bassiana* strain GHA)

Crop and/or situation (a)	Member State or Country	Product name	F G or I (b)	Pests or Group of pests controlled (c)	Preparation		Application				Application rate per treatment			PHI (days) (l)	Remarks (m)
					Type (d-f)	Conc. of MPCA (i)	Method Kind (f-h)	Growth stage & season (j)	Number min max (k)	Interval between applications (min)	Kg MPCA/hL min min	water L/ha min max	Kg MPCA/ha Viable granules MPCA/ha min max		
Ornamentals	Northern and Southern Europe	BotaniGard 22 WP	G, I	Sucking insects	WP	220 g/kg 4.4 x 10 ¹³ CFU/kg	High volume Spraying	Maturity (BBCH89)	3 - 5	5-7 days	0.14 - 4.84 x 10 ¹² CFU	500 - 2000	0.014 – 0.121 // 0.28-2.42 x 10 ¹³ CFU	0	MPCP 0.0625-0.55 Kg/ha
Tomatoes, Cucumbers	Northern and Southern Europe	BotaniGard 22 WP	G, I	Sucking insects	WP	220 g/kg 4.4 x 10 ¹³ CFU/kg	High volume Spraying	Ripeness (BBCH89)	3 - 5	5-7 days	0.14 - 4.84 x 10 ¹² CFU	500 - 2000	0.014 – 0.121 // 0.28-2.42 x 10 ¹³ CFU	0	MPCP 0.0625-0.55 Kg/ha

- (a) For crops, the EU and codex classifications (both) should be used; where relevant, the situation should be described (e.g. fumigation of a structure)
- (b) Outdoor or field use (F), glasshouse application (G) or indoor application (I)
- (c) e.g. biting and sucking insects, soil born insects, foliar fungi, weeds
- (d) e.g. wettable powder (WP), emusifiable concentrate (EC), granule (GR)
- (e) GCPF codes - Crop Life Technical Monograph No 2, 1989
- (f) All abbreviations used must be explained
- (g) Method, e.g. high volume spraying, spreading, dusting, drench
- (h) Kind, e.g. overall, broadcast, aerial spraying, row, individual plant, between the plant - type of equipment used must be indicated
- (i) viable granules = colony forming units and g/kg or g/L
- (j) Growth stage at last treatment (BBCH Monograph, growth stages of Plants, 1997, Blackwell ISBN 3-8263-3152-4), including where relevant, information on season at time of application
- (k) Indicate the minimum and maximum number of application possible under practical conditions of use
- (l) PHI - minimum pre-harvest interval
- (m) Remarks may include: Extent of use/economic importance/restrictions

Classification and proposed labelling (Symbol, Indication of danger, Risk phrases, Safety phrases)

with regard to physical/chemical data:	none
with regard to toxicological data:	R42 – May cause sensitisation by inhalation R43 – May cause sensitisation by skin contact
with regard fate and behaviour:	Not applicable
with regard to ecotoxicological data:	Not applicable

Chapter 2: Methods of analysis

Analytical methods for the microorganism (OECD data point IIM 4.2, 4.3 and IIM 5.3)

manufactured micro-organism (principal of method):	Analytical methods for the identification of the technical <i>B. bassiana</i> strain GHA include the following: - Characterisation by microscopic examination of the visual morphology - Restriction Fragment Length Polymorphism (RFLP) profile analysis of chromosomal DNA (cDNA) and Random Amplified Polymorphic DNA analysis (RAPD) Determination of fungal spore-count and spore viability by haemocytometer counts and <i>in vitro</i> germination assay. Open for validation
Impurities and contaminating micro-organisms in manufactured material (principal of method):	Enumeration of microbial contaminants (bacteria and fungi), determining the presence/absence of enteric bacteria such as <i>Shigella</i>, <i>Salmonella</i> and <i>Vibrio spp.</i> by plating out on selective media. The method for beauvericin is not sufficiently validated. Moisture analyser balance. Open for contaminating microorganisms and toxins
Microbial Pest control product (principle of method):	Analytical methods for the formulation are as used for the active substance.

Analytical methods for residues (viable and non-viable) in exposed compartments and organisms (OECD data point IIM 4.5)

of the active micro-organism (principle of method)	Validated methods are not required and not available. Nonetheless one method is validated for water with an LOQ of 1×10^6 CFU/L.
Of relevant metabolites (principle of method):	None relevant

Chapter 3: Impact on Human and Animal Health

Medical data: (including medical surveillance on manufacturing plant personnel) (OECD data point IIM 5.1, 5.2)	No adverse effects in employees who had been in occupational contact with strain GHA. For <i>Beauveria bassiana</i> in general, few cases of local (eye) and systemic infections were reported with the latter occurring only in immunocompromised or severely ill patients. Evidence of allergenicity by respiratory or skin contact.
Sensitisation: (OECD data point IIM 5.3.1 & IIM 7.1.6)	Microorganisms may have the potential to provoke sensitising reactions. Evidence of allergenicity by inhalation in laboratory animals and humans and indications of allergic skin reactions in humans. Buehler test in Guinea pigs negative for the micro-organism but positive for the formulation BotaniGard 22 WP.
Acute oral infectivity, toxicity and pathogenicity: (OECD data point IIM 5.3.2 & IIM 7.1.1)	LD ₅₀ > 1 x 10 ⁸ CFU/animal in rats; no evidence of toxicity, pathogenicity or infectivity; rapid clearance (by day 3 post-dosing)
Acute intratracheal/inhalation toxicity, pathogenicity and infectivity: (OECD data point IIM 5.3.3 & IIM 7.1.3)	LD ₅₀ > 1 x 10 ⁸ CFU/animal in rats after intratracheal application; no evidence of infectivity but local effects on the lungs (inflammation/immune reaction, organ weight increase), transient reduction in body weight gain; clearance complete by day 7 post-dosing
Acute intravenous/intraperitoneal toxicity, pathogenicity and infectivity: (OECD data point IIM 5.3.4 & IIM 7.1.2)	LD ₅₀ > 1 x 10 ⁷ CFU in rats; no evidence of toxicity, pathogenicity or infectivity; rapid clearance (by day 3 post-dosing)
Genotoxicity: (OECD data point IIM 5.3.5)	Negative in the Ames test
Cell culture study: (OECD data point IIM 5.3.6)	Not available, not required
Information on short-term toxicity and pathogenicity: (OECD data point IIM 5.3.7)	Not available, not required
Dermal toxicity: (OECD data point IIM 7.1.2)	Acute dermal study in rabbits: LD ₅₀ > 1.6x 10 ¹¹ CFU; no systemic effects but signs of slight but persisting local irritation.
Specific-toxicity, pathogenicity and infectivity: (OECD data point IIM 5.5)	Acute dermal study in rabbits: LD ₅₀ > 1.6x 10 ¹¹ CFU; no systemic effects but signs of slight but persisting local irritation; Certain potential of the micro-organism (strain not specified) to damage the cornea confirmed in rabbits; Long-term inhalative exposure to <i>B. bassiana</i> (strain not specified) caused allergic lung reactions in rats and mice.

Genotoxicity – *in vivo* studies in germ cells:
(OECD data point IIM 5.5.3)

Not available, not required

Metabolites

Toxicological studies with strain GHA including a beauvericin and bassianolide contamination each with of approx. 50 mg/kg indicated no additional adverse effects after oral, dermal or ip administration. Effects seen after intratracheal administration are unlikely to be due to beauvericin and bassianolide.

Beauvericin was not mutagenic in an Ames test.

Exposure to (the sum of) the metabolites should stay below 1.5 µg/d (threshold of toxicological concern).

Reference values

AOEL

There is no need to set an AOEL as the microorganism was not shown to be pathogenic or infective based on the available data and studies

ARfD

There is no need to set an ARfD as the microorganism was not shown to be pathogenic or infective based on the available data and studies

ADI

There is no need to set an ADI as the microorganism was not shown to be pathogenic or infective based on the available data and studies

Exposure (operator, workers, bystander, consumer):

(OECD data point IIM 6.1, IIM 7.2, 7.3 and 8.0)

Operator

Not finalised (lack of information on toxin/metabolites production)

Bystander

Bystander exposure is not relevant for the representative uses (greenhouse).

Worker

Not finalised (lack of information on toxin/metabolites production).

Chapter 4: Residues

Open for toxins/secondary metabolites

Chapter 5: Fate and Behaviour in the Environment (OECD IIM 7 & IIM 9)

Persistence and multiplication in soil, water and air:

In Soil:

Data gap

The available data on measurement or observation of persistence and multiplication in soil were insufficient.

In Water:

In an aquatic environment spores will be subject to sedimentation and can be expected to degrade rapidly in natural water bodies. The conidia did not germinate in different tested aqueous systems (distilled water, and aqueous suspensions at pH 5, 7 and 9, or in metal ion solutions in the presence of bacteria).

In Air:

In the absence of a specific host insect, viability of conidia of *B. bassiana* in the air or epigeal habitats is mainly affected by the UV-B portion of the solar spectrum. On the basis of the persistence data, significant degradation of *B. bassiana* conidia on leaves under field conditions may occur in less than 2 days.

Mobility

In soil:

Conidia of *B. bassiana* are not very mobile in soil and generally remain on the surface of the soil. No groundwater risk assessment is necessary since *B. bassiana* is neither pathogenic nor toxic to humans.

In water:

The globose or ovoid shape of conidia is not suitable for efficient transport in water.

In air:

Beauveria bassiana is dry, of small size and it is produced in powdery clusters. Therefore, these types of conidia can be easily transported through the air. Also transmission of fungal spores through the air by insects is possible.

However, in the absence of specific host insects, conidia of *Beauveria bassiana* will not persist in the sun-exposed phylloplane in significant amount for more than 2 days.

Chapter 6: Effects on non-target organisms (OECD IIM 8 & IIM 10)

Effects on birds (OECD IIM 8.1 & IIM 10.1)

Application rate (kg MPCA/ha)	Crop	Category (e.g. insectivorous bird)	Time-scale	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
0.121 (corresponding to 2.42×10^{13} CFU/ha)	Ornamentals, Tomatoes, Cucumbers	Insectivorous bird	5 days dosing / 18 days observation	LD ₅₀ (<i>Falco sparverius</i>) > 2.5×10^{10} conidia/kg body weight (no evidence of pathogenicity)

Effects on Mammals

Application rate (kg MPCA/ha)	Crop	Category (e.g. insectivorous bird)	Time-scale	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
tomatoes and cucumbers, indoor	ornamentals, tomatoes and cucumbers, indoor	rat	acute	LD ₅₀ > 5000 mg BotaniGard 22WP/ kg bw corresponding to > 2.2×10^{14} CFU/ kg bw (product containing 4.4×10^{13} CFU/kg)
		rat	acute	<i>Beauveria bassiana</i> strain GHA was not toxic, pathogenic, or infective to male and female rats at 1.05×10^8 CFU and 1×10^8 CFU respectively.

Effects on aquatic organisms (OECD IIM 8.2, 8.3 & IIM 10.2)

Group	Test substance	Time-scale	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
Laboratory tests			
Fish species: <i>Pimephales promelas</i>	<i>Beauveria bassiana</i> GHA	31 days	NOEC < 7.5×10^8 CFU/L (effects on growth < 25 %; no evidence of infectivity or pathogenicity)
Invertebrate species: <i>Daphnia magna</i>	<i>Beauveria bassiana</i> GHA	21 days	NOEC = 4.7×10^8 CFU/L (sublethal effects; no evidence of infectivity or pathogenicity)

Effects on algae: (growth, growth rate, capacity to recover) (OECD IIM 8.4 & IIM 10.2)	<i>Selenastrum capricornutum</i> , 3 days, static, nominal E _b C ₅₀ 114000 µg/L (0.98×10^{13} CFU/L) E _r C ₅₀ 237000 µg/L (2.04×10^{13} CFU/L) NOEC 75000 µg/L (2.64×10^{13} CFU/L) (all parameters)
Effects on aquatic plants (OECD IIM 8.5 & IIM 10.2)	not performed, not required

Effects on bees (OECD IIM 8.7 & IIIM 10.3)

Application rate (kg MPCA/ha)	Crop	Route	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
No data available			
Study under practical conditions (honey bees)			
Laboratory Test (bumblebees)			

Effects on terrestrial arthropods other than bees (OECD IIM 8.8 & IIIM 10.4)

Species	Stage	Test Substance	Dose kg MPCA/ha)	Toxicity, infectivity and pathogenicity (endpoint, value or other description of effects)
Laboratory Tests				
<i>Aphidius rhopalosiphi</i>	adult	BotaniGar d 22 WP BotaniGar d ES	0.947 (1.33 x 10 ¹⁴ CFU/ha) 1.695	Mortality: -5 % Reproduction: 20 % Mortality: -5 % Reproduction: 20 %
<i>Typhlodromus pyri</i>	protonymph	BotaniGar d 22 WP BotaniGar d ES	0.947 (1.33 x 10 ¹⁴ CFU/ha) 1.695	Mortality: 4 % Reproduction: 43 % Mortality: -4 % Reproduction: 34 %
<i>Orius laevigatus</i>	nymph	BotaniGar d 22 WP BotaniGar d ES	0.952 (1.33 x 10 ¹⁴ CFU/ha) 1.693	Mortality: 3 % Reproduction: 23 % Hatching succ: 24 % Mortality: 11 % Reproduction: -7 % Hatching succ: 29 %
<i>Pardosa spec.</i>	adult	BotaniGar d 22 WP BotaniGar d ES	No sufficient information No sufficient information	Mortality: 80 % Mortality: 45 %
Further information				
<i>Xylocoris flavipes</i>	nymphs	<i>Beauveria bassiana</i> GHA	Filter paper: 2.5 x 10 ¹³ , 2.5 x 10 ¹⁴ , 2.5 x 10 ¹⁵ CFU/ha	0, 16, 41 % infection after 10 days, respectively.
<i>Tenebrio molitor</i>	adult	<i>Beauveria bassiana</i> GHA	2 x 10 ⁴ CFU, 2.4 x 10 ⁸ CFU	No significant mortality (100 % mortality of target organism <i>Melanoplus sanguinipes</i> at 2 x 10 ⁴ conidia/ha)

Effects on other terrestrial invertebrates (OECD IIM 8.9, 8.9.1 & IIM 10.5)

Toxicity, infectivity and pathogenicity: (endpoint, value or other description of effects)	LC₅₀ (Mortality) : > 1000 mg <i>Beauveria bassiana</i> strain GHA /kg artificial soil > 8.6 x 10¹⁰ CFU/kg artificial soil (<i>Eisenia foetida</i>)
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Effects on non-target soil micro-organisms (OECD IIM 8.10 & IIM 10.6)

No studies submitted

Additional studies (OECD IIM 8.11 & IIM 10.7)

No data available for the biological methods of sewage treatment plants, data gap identified.
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ABBREVIATIONS

1/n	slope of Freundlich isotherm
λ	wavelength
ε	decadic molar extinction coefficient
°C	degree Celsius (centigrade)
μg	microgram
μm	micrometer (micron)
a.s.	active substance
AChE	acetylcholinesterase
ADE	actual dermal exposure
ADI	acceptable daily intake
AF	assessment factor
AOEL	acceptable operator exposure level
AP	alkaline phosphatase
AR	applied radioactivity
ARfD	acute reference dose
AST	aspartate aminotransferase (SGOT)
AV	avoidance factor
BCF	bioconcentration factor
BUN	blood urea nitrogen
bw	body weight
CAS	Chemical Abstracts Service
CFU	colony forming units
ChE	cholinesterase
CI	confidence interval
CIPAC	Collaborative International Pesticides Analytical Council Limited
CL	confidence limits
cm	centimetre
d	day
DAA	days after application
DAR	draft assessment report
DAT	days after treatment
DM	dry matter
DT ₅₀	period required for 50 percent disappearance (define method of estimation)
DT ₉₀	period required for 90 percent disappearance (define method of estimation)
dw	dry weight
EbC ₅₀	effective concentration (biomass)
EC ₅₀	effective concentration
ECHA	European Chemical Agency
EEC	European Economic Community
EINECS	European Inventory of Existing Commercial Chemical Substances
ELINCS	European List of New Chemical Substances
EMDI	estimated maximum daily intake
ER ₅₀	emergence rate/effective rate, median
ErC ₅₀	effective concentration (growth rate)
EU	European Union
EUROPOEM	European Predictive Operator Exposure Model
f(twa)	time weighted average factor
FAO	Food and Agriculture Organisation of the United Nations
fd	feed
FIR	Food intake rate
FOB	functional observation battery
FOCUS	Forum for the Co-ordination of Pesticide Fate Models and their Use
g	gram

GAP	good agricultural practice
GC	gas chromatography
GCPF	Global Crop Protection Federation (formerly known as GIFAP)
GGT	gamma glutamyl transferase
GM	geometric mean
GS	growth stage
GSH	glutathion
h	hour(s)
ha	hectare
Hb	haemoglobin
Hct	haematocrit
hL	hectolitre
HPLC	high pressure liquid chromatography or high performance liquid chromatography
HPLC-MS	high pressure liquid chromatography – mass spectrometry
HQ	hazard quotient
IEDI	international estimated daily intake
UESTI	international estimated short-term intake
ISO	International Organisation for Standardisation
IUPAC	International Union of Pure and Applied Chemistry
JMPR	Joint Meeting on the FAO Panel of Experts on Pesticide Residues in Food and the Environment and the WHO Expert Group on Pesticide Residues (Joint Meeting on Pesticide Residues)
K _{doc}	organic carbon linear adsorption coefficient
kg	kilogram
K _{Foc}	Freundlich organic carbon adsorption coefficient
L	litre
LC	liquid chromatography
LC ₅₀	lethal concentration, median
LC-MS	liquid chromatography-mass spectrometry
LC-MS-MS	liquid chromatography with tandem mass spectrometry
LD ₅₀	lethal dose, median; dosis letalis media
LDH	lactate dehydrogenase
LOAEL	lowest observable adverse effect level
LOD	limit of detection
LOQ	limit of quantification (determination)
m	metre
M/L	mixing and loading
MAF	multiple application factor
MATC	maximum allowable toxicant concentration
MCH	mean corpuscular haemoglobin
MCHC	mean corpuscular haemoglobin concentration
MCV	mean corpuscular volume
mg	milligram
mL	millilitre
mm	millimetre
mN	milli-newton
MPCA	microbial pest control agent
MPCP	microbial pest control product
MRL	maximum residue limit or level
MS	mass spectrometry
MSDS	material safety data sheet
MTD	maximum tolerated dose
MWHC	maximum water holding capacity
NESTI	national estimated short-term intake

ng	nanogram
NOAEC	no observed adverse effect concentration
NOAEL	no observed adverse effect level
NOEC	no observed effect concentration
NOEL	no observed effect level
OD	oil dispersion
OECD	Organisation for Economic Co-operation and Development
OM	organic matter content
Pa	pascal
PD	proportion of different food types
PEC	predicted environmental concentration
PEC _{air}	predicted environmental concentration in air
PEC _{gw}	predicted environmental concentration in ground water
PEC _{sed}	predicted environmental concentration in sediment
PEC _{soil}	predicted environmental concentration in soil
PEC _{sw}	predicted environmental concentration in surface water
pH	pH-value
PHED	pesticide handler's exposure data
PHI	pre-harvest interval
PIE	potential inhalation exposure
pK _a	negative logarithm (to the base 10) of the dissociation constant
P _{ow}	partition coefficient between <i>n</i> -octanol and water
PPE	personal protective equipment
ppm	parts per million (10 ⁻⁶)
ppp	plant protection product
PT	proportion of diet obtained in the treated area
PTT	partial thromboplastin time
QSAR	quantitative structure-activity relationship
r ²	coefficient of determination
RAPD	random amplification of polymorphic DNA
REACH	Registration, Evaluation, Authorisation of CHemicals
RPE	respiratory protective equipment
RUD	residue per unit dose
SC	suspension concentrate
SD	standard deviation
SFO	single first-order
SSD	species sensitivity distribution
STMR	supervised trials median residue
t _{1/2}	half-life (define method of estimation)
TER	toxicity exposure ratio
TER _A	toxicity exposure ratio for acute exposure
TER _{LT}	toxicity exposure ratio following chronic exposure
TER _{ST}	toxicity exposure ratio following repeated exposure
TK	technical concentrate
TLV	threshold limit value
TMDI	theoretical maximum daily intake
TRR	total radioactive residue
TSH	thyroid stimulating hormone (thyrotropin)
TWA	time weighted average
UDS	unscheduled DNA synthesis
UV	ultraviolet
W/S	water/sediment
w/v	weight per volume
w/w	weight per weight
WBC	white blood cell

WG	water dispersible granule
WHO	World Health Organisation
wk	week
yr	year